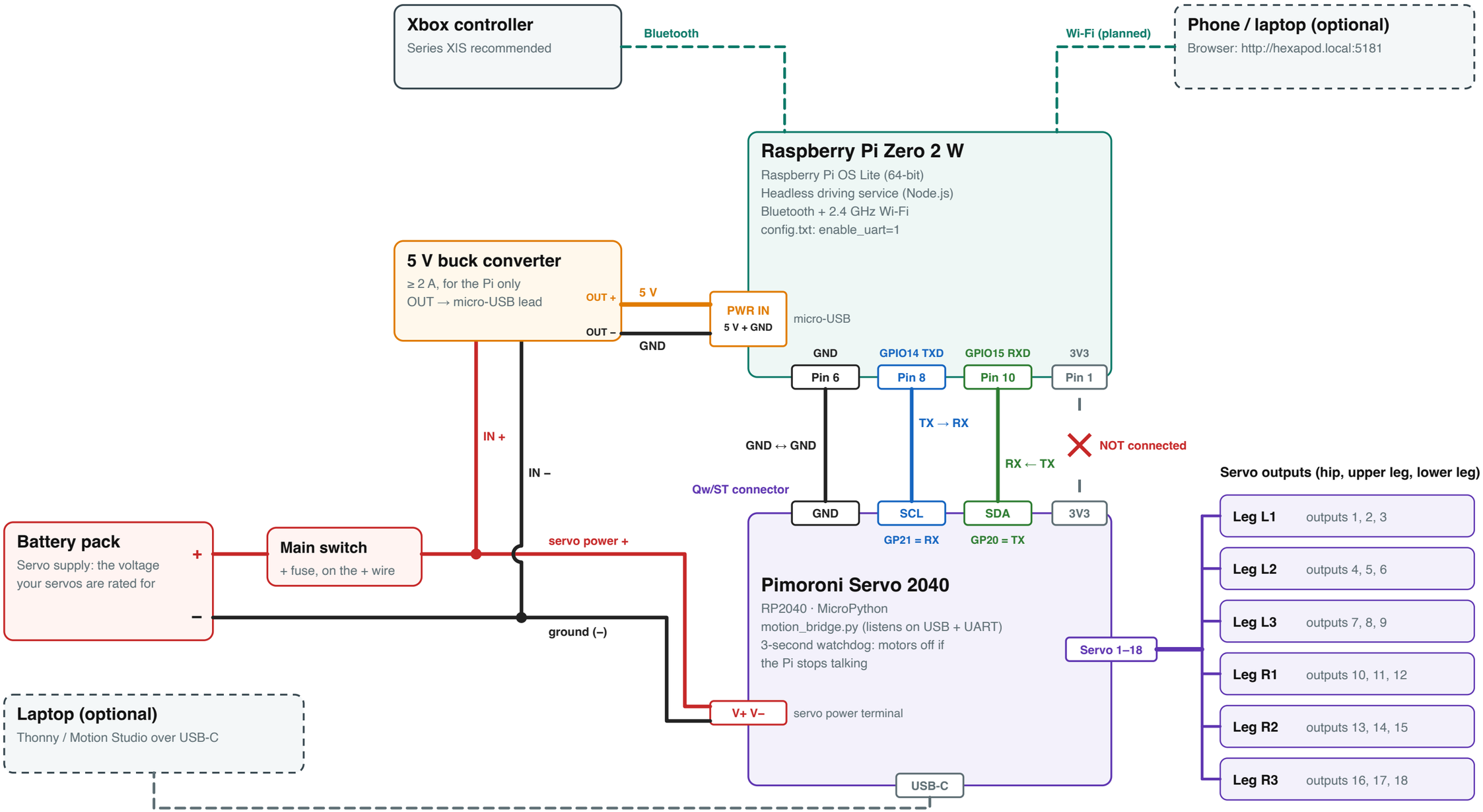
Hexapod — Raspberry Pi Zero 2 W wiring schematic

Xbox controller (Bluetooth) → Pi Zero 2 W → UART on the Servo 2040 SDA/SCL pins → 18 servos



Before wiring

1. VERIFY on the Servo 2040 pinout that SDA = GP20 and SCL = GP21 (UART1 TX/RX).
2. Cross the wires: Pi TX → Servo RX, Pi RX ← Servo TX. Join GND. Do NOT join 3V3.
3. Both boards use 3.3 V logic: no level shifter. Keep UART wires under 20–30 cm.
4. Power the Pi from its own 5 V regulator (5 V and GND), never from the servo rail directly.
5. Common ground: battery –, buck converter, Pi and Servo 2040 GND are all joined.
6. Switch the servo power off before connecting or disconnecting anything.

Legend — Battery / servo power + · Ground (–) · 5 V for the Pi · UART: Pi → Servo 2040 · UART: Servo 2040 → Pi · Wireless

Hexapod Pi Zero 2 W
Wiring schematic · Rev B
4 November 2026

Status: planned, not yet built
Pin labels to be verified